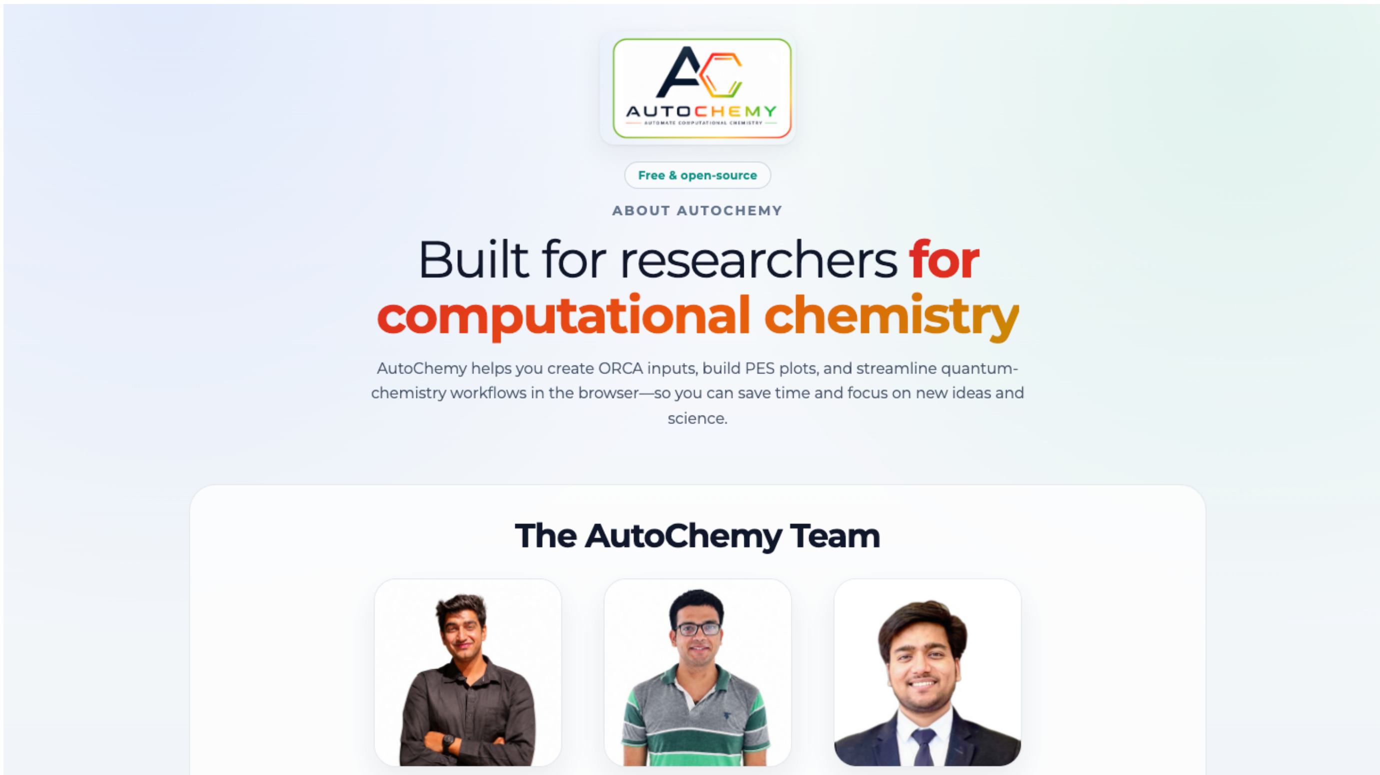
AUTOCHEMY

Quantum-Chemistry Workflows, Made Visual

A concise overview of AutoChemy's tools for ORCA input creation, calculation setup, xTB screening, reaction-pathway management, potential-energy analysis, and live browser access.

Free and open source | Built for computational chemistry researchers | autochemy.xyz | GitHub

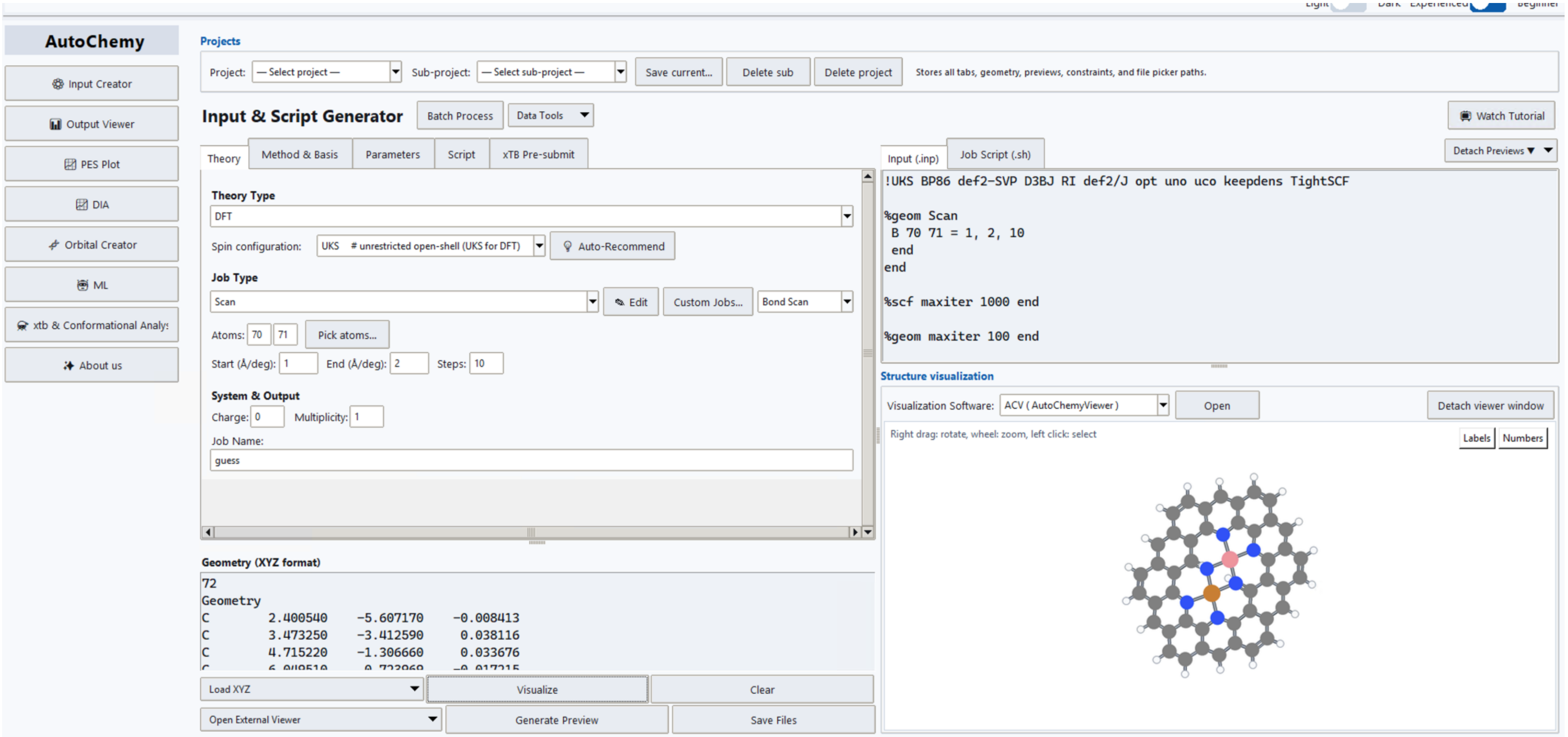


Create inputs. Run and inspect calculations. Build clear mechanistic figures.

FEATURE 01

ORCA Input & HPC Script Generation

Build ORCA calculations from guided controls while AutoChemy generates both the input file and a reusable shell script beside the molecular structure.



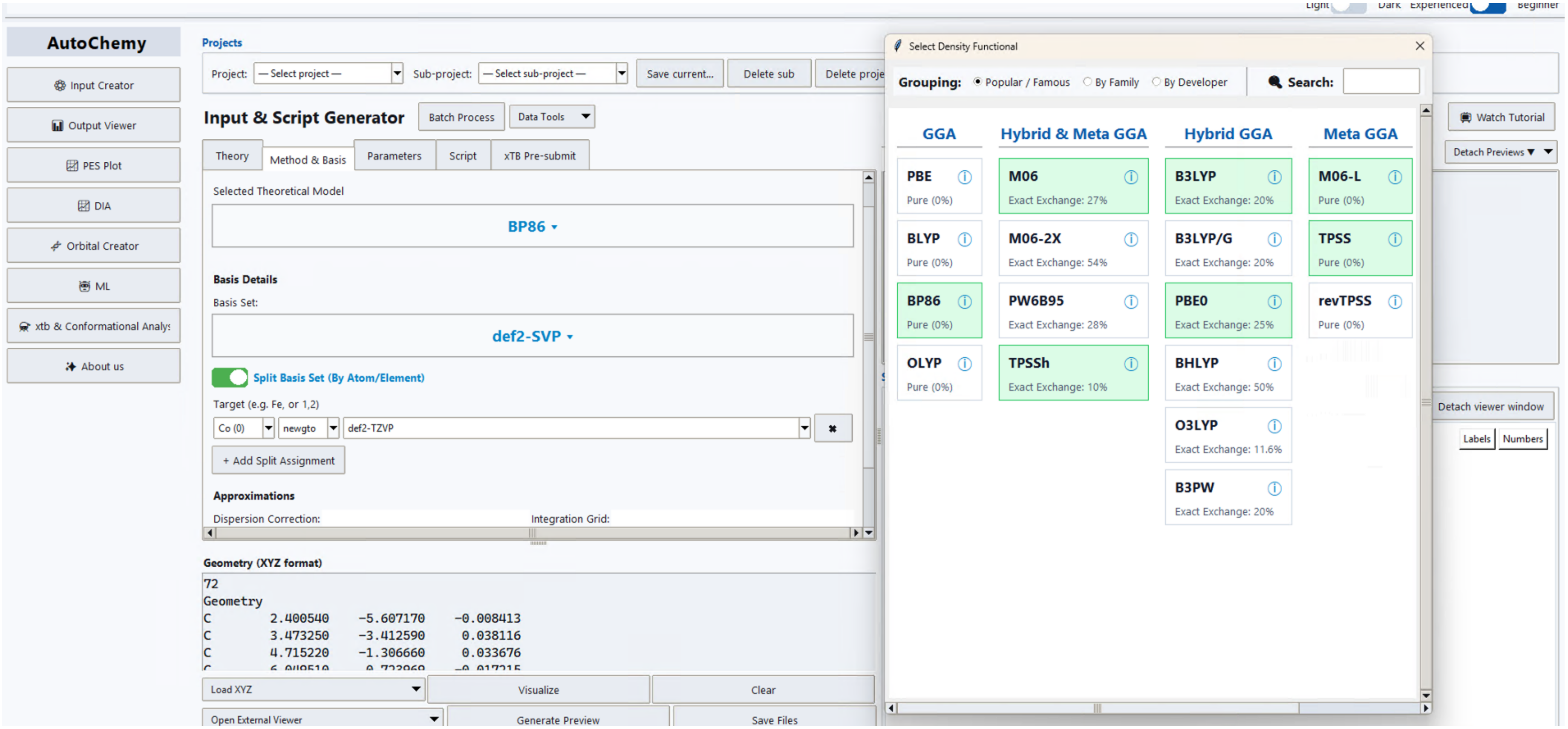
Guided calculation setup: Configure theory, spin, job type, scan coordinates, geometry, convergence settings, and advanced ORCA keywords in one workspace.

Local or HPC: Generate ORCA .inp and shell .sh files for a workstation or adapt the script for an HPC scheduler and cluster environment.

FEATURE 02

Functional & Basis-Set Selection

Choose density functionals and basis sets from organized scientific libraries instead of manually recalling and typing every ORCA keyword.



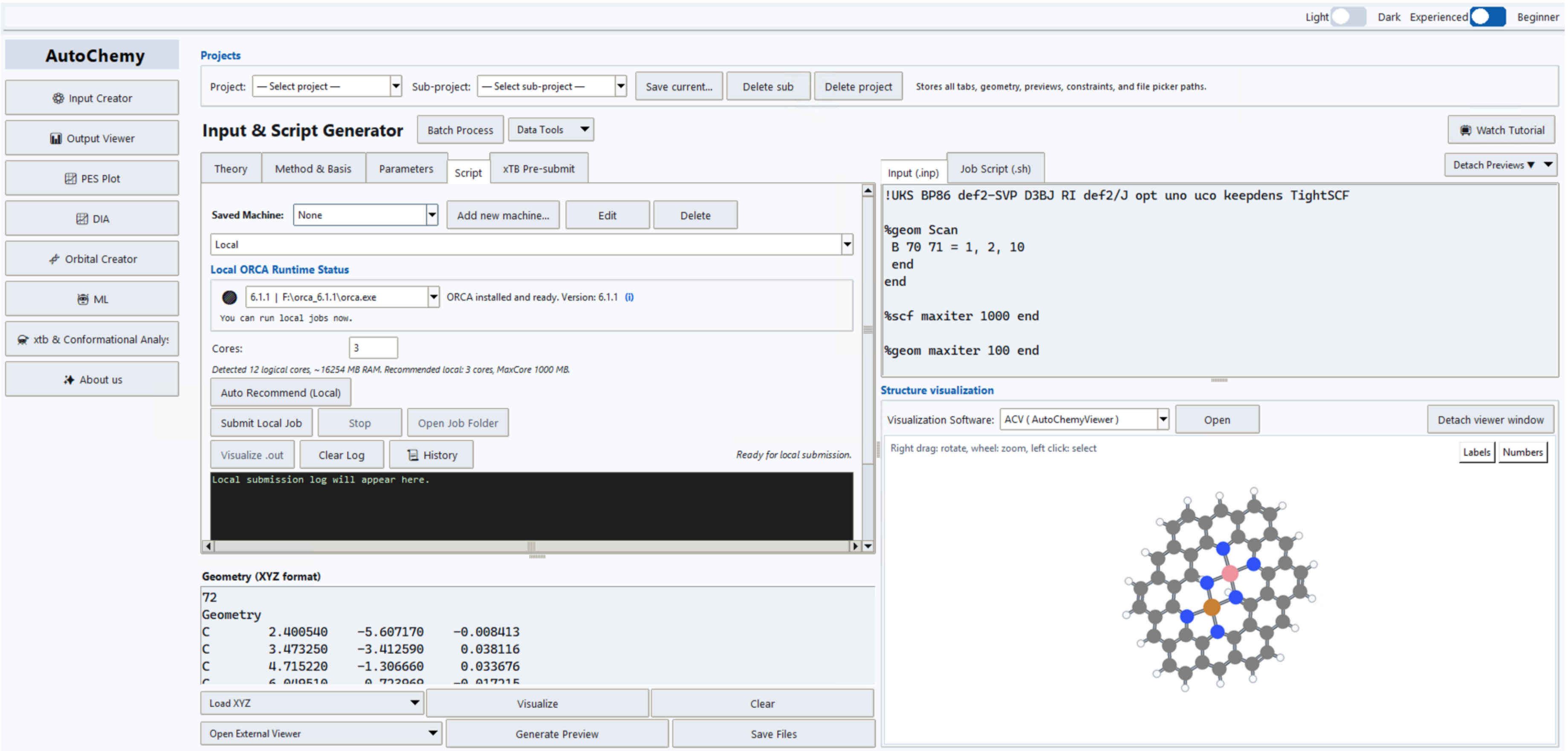
Searchable functional library: Browse popular functionals or group them by family or developer, with useful method information shown directly in the selector.

Flexible basis assignment: Select a main basis set and apply split basis sets to specific atoms or elements for tailored computational protocols.

FEATURE 03

Run ORCA Locally or Use Machine Profiles

Detect a local ORCA installation, configure computational resources, submit calculations directly, or preserve reusable machine settings for scripted HPC workflows.



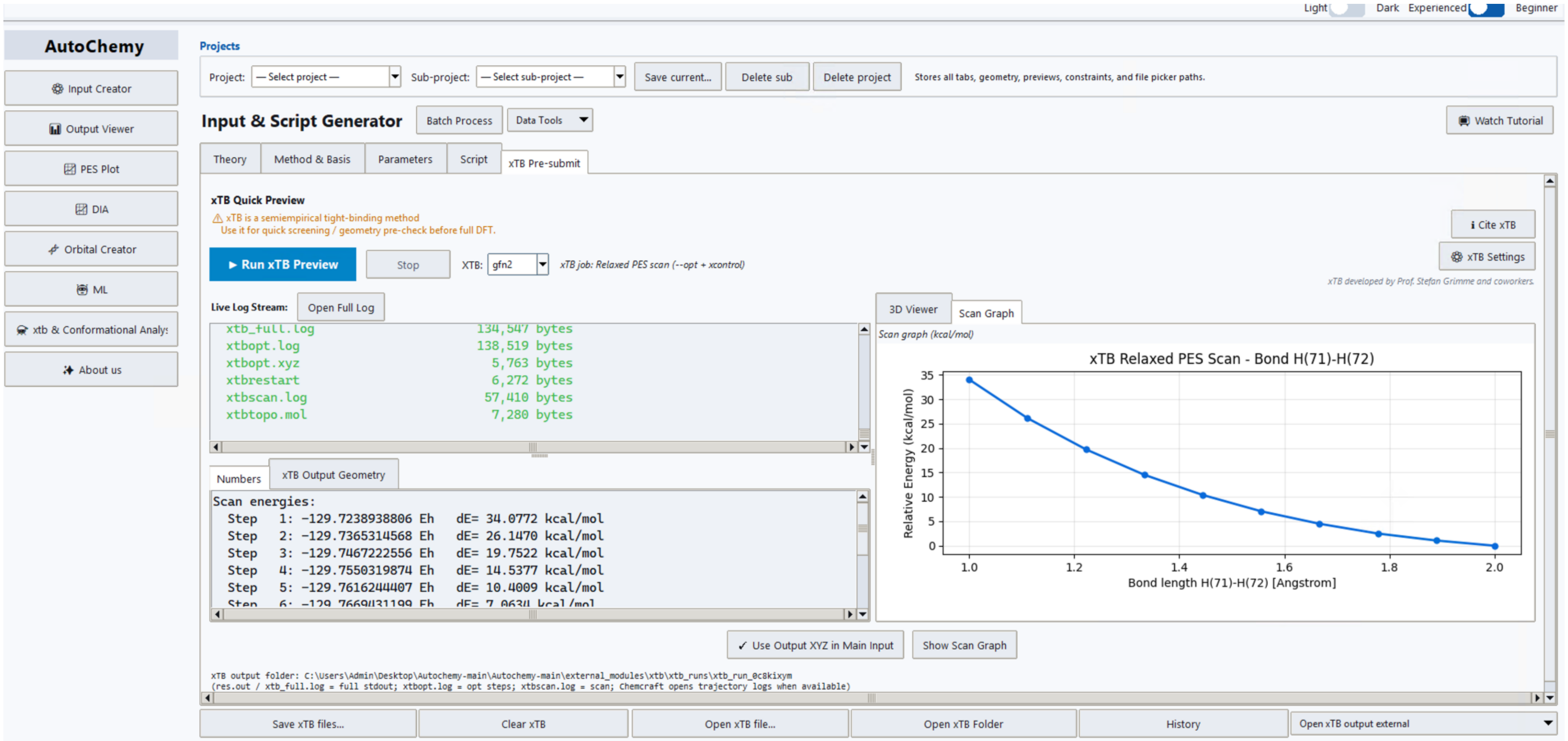
Run on your computer: AutoChemy displays the detected ORCA version, available cores, RAM context, and recommended local settings before submission.

Scale to HPC: Saved machine profiles and generated job scripts make repeated cluster submissions more consistent and easier to reproduce.

FEATURE 04

Fast xTB Pre-Screening

Run a quick Lates g-xtb/GFN2-xTB preview before full DFT, inspect the live calculation log, and view a relaxed scan as both numerical energies and a graph.



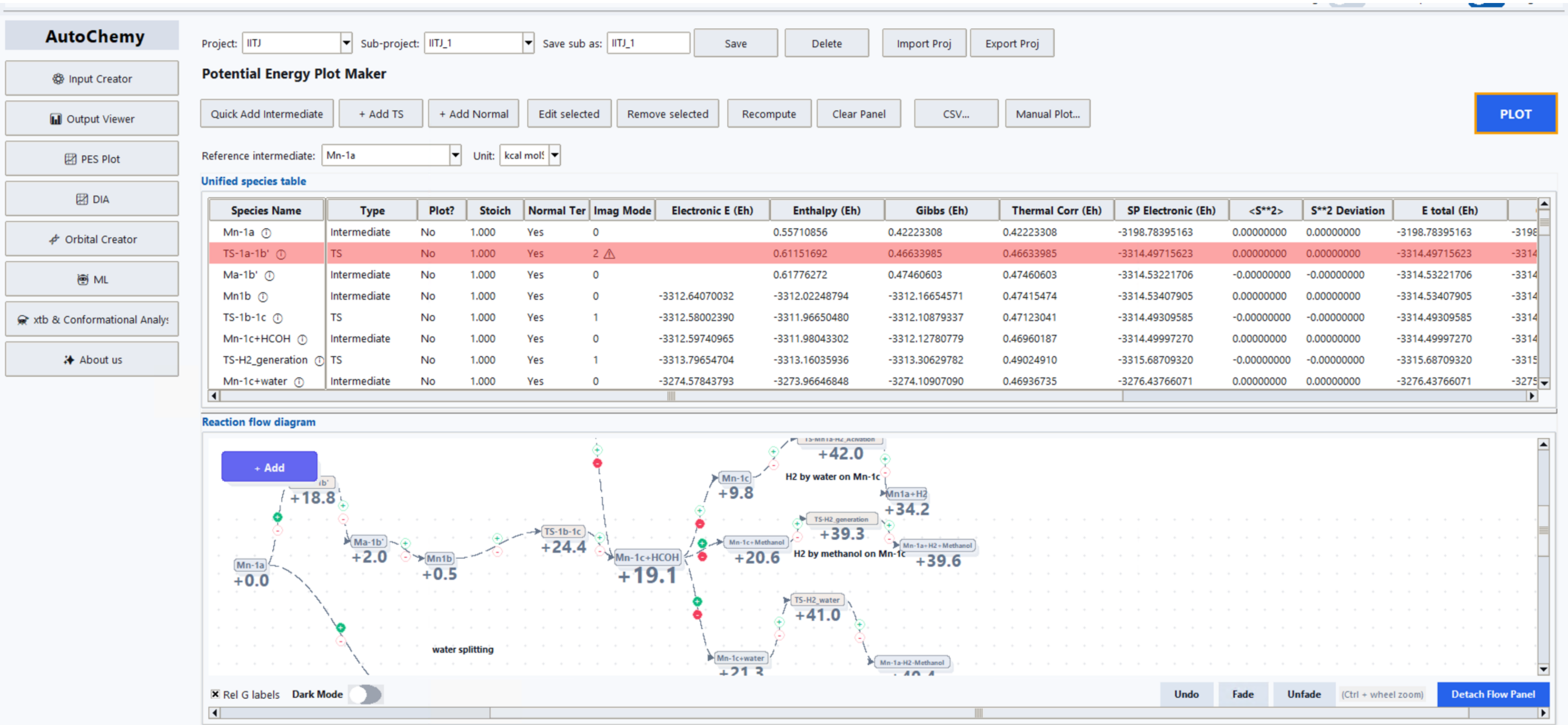
Early insight: Rapid scans help identify useful geometries, trends, and problematic regions before a more expensive calculation.

Direct reuse: The resulting xTB geometry can be transferred back into the main ORCA input workflow.

FEATURE 05

Reaction-Pathway Workspace

Collect intermediates and transition states in a unified energy table, then arrange them as an editable reaction-flow diagram.



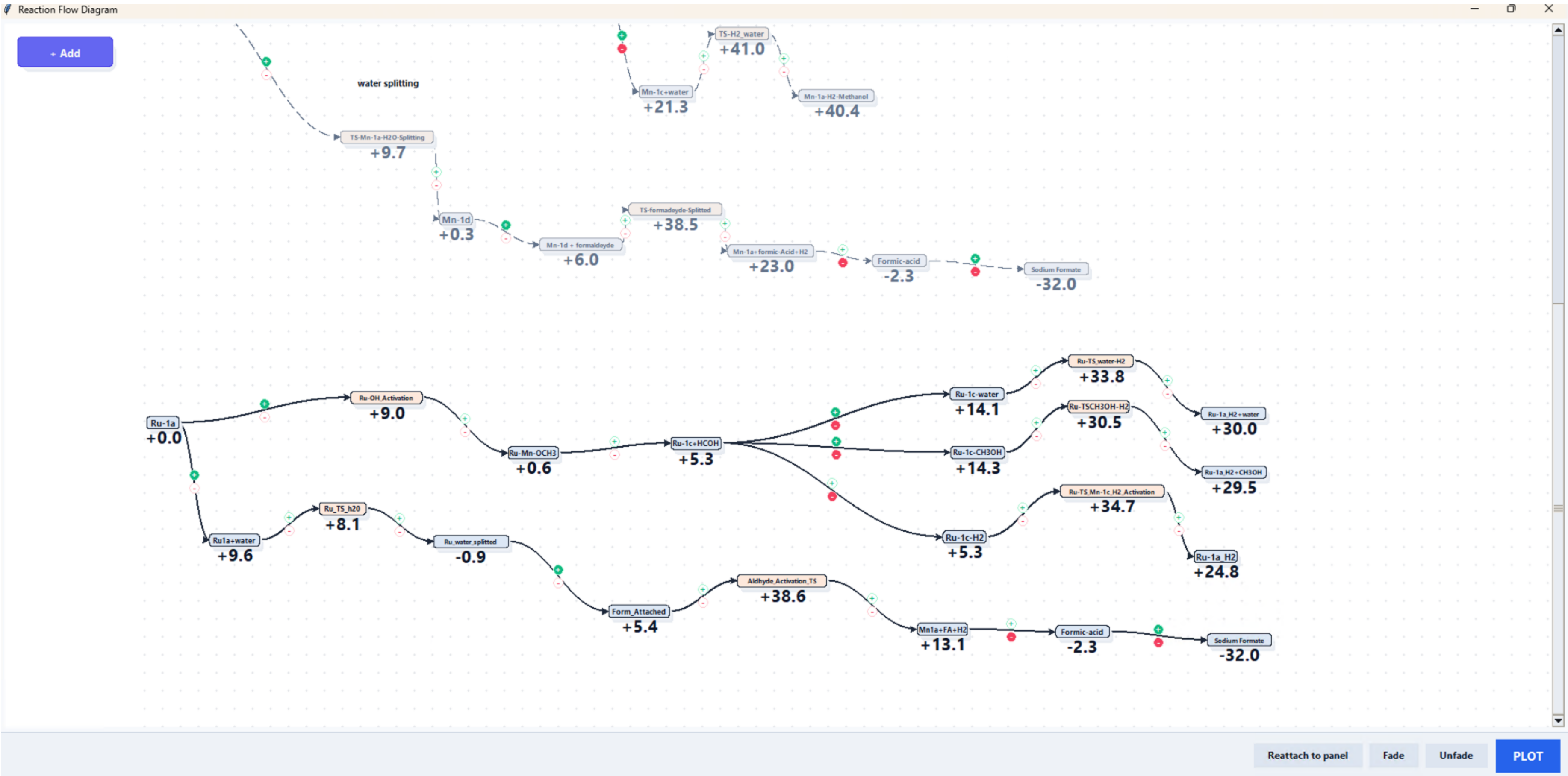
Data in context: Electronic, thermal, enthalpy, Gibbs, and spin-related values remain connected to each species.

Mechanistic organization: Branches and competing pathways can be assembled visually before producing the final PES figure.

FEATURE 06

Detachable Reaction-Flow Canvas

Open the pathway diagram in a large independent window to inspect complex catalytic networks and compare multiple mechanistic branches.



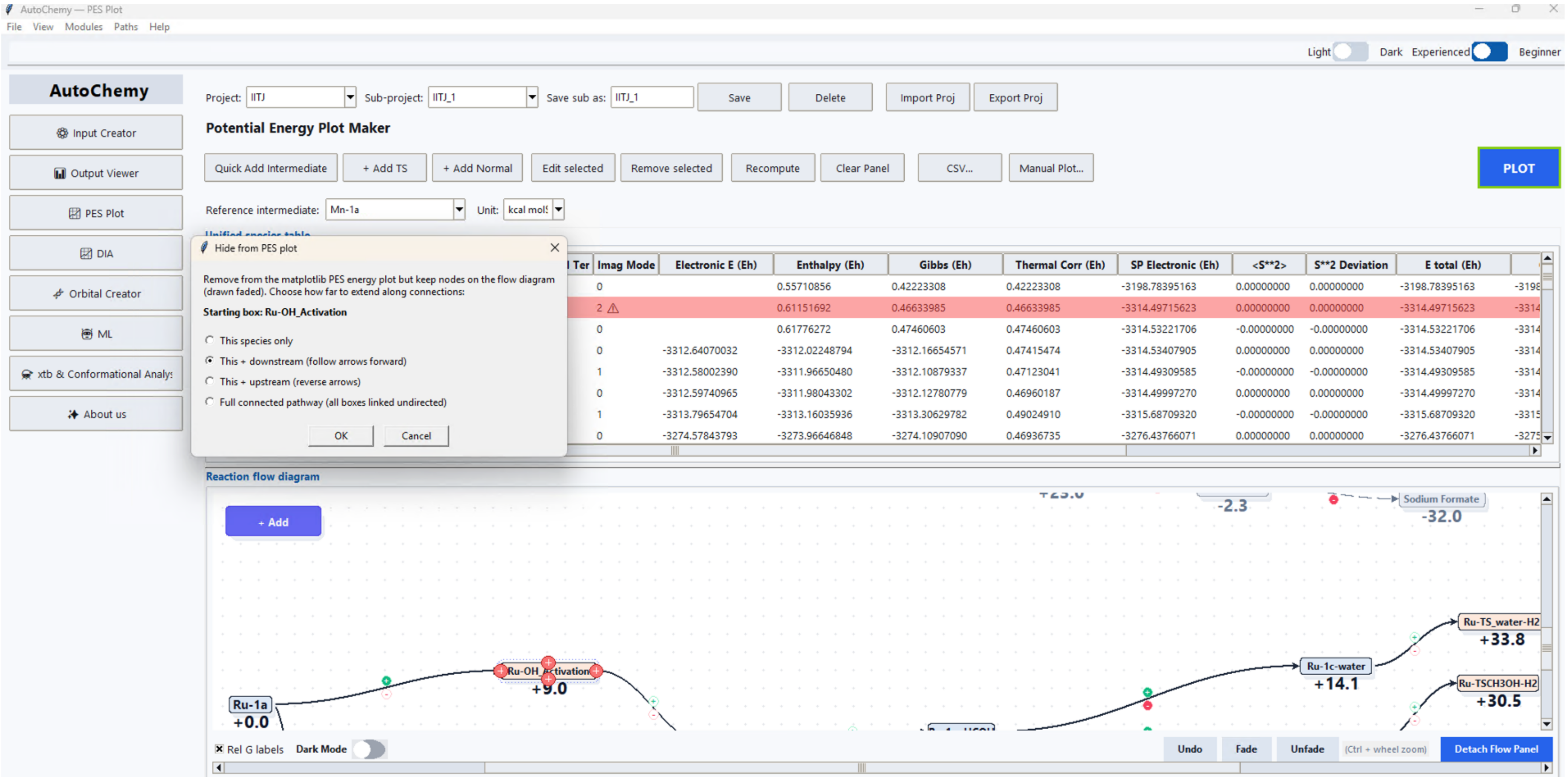
Large-network visibility: A spacious canvas keeps long and branched pathways readable during analysis.

Interactive editing: Species and connections can be extended, faded, restored, and reorganized as the mechanism develops.

FEATURE 07

Selective Pathway Control

Choose exactly which species or connected pathway should appear in the final PES plot while retaining the complete reaction network in the workspace.



Focused figures: Hide one species, a downstream branch, an upstream branch, or an entire connected pathway.

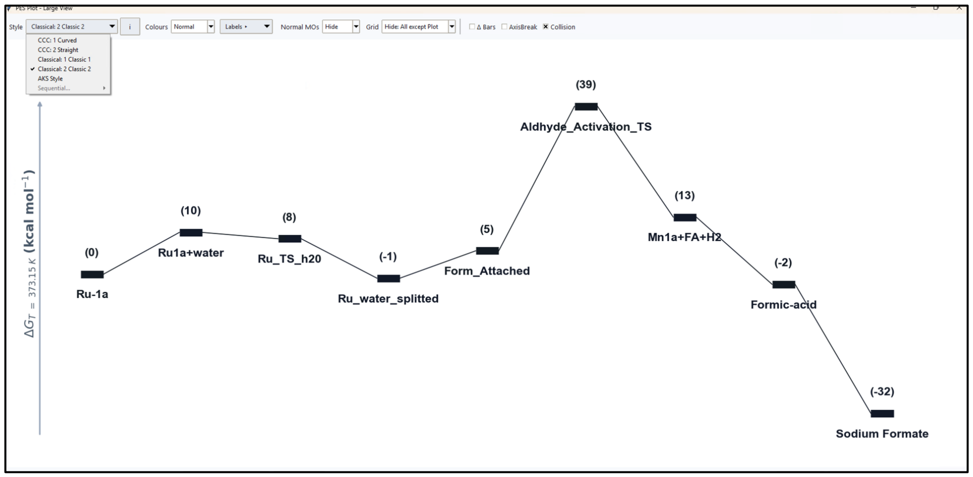
Non-destructive analysis: Nodes remain available in the reaction-flow diagram, so alternate mechanisms are not lost.

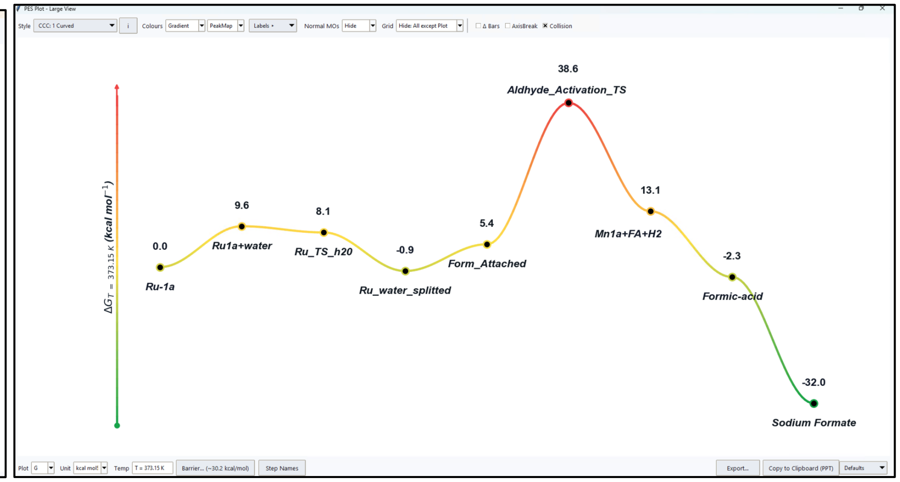
FEATURE 08

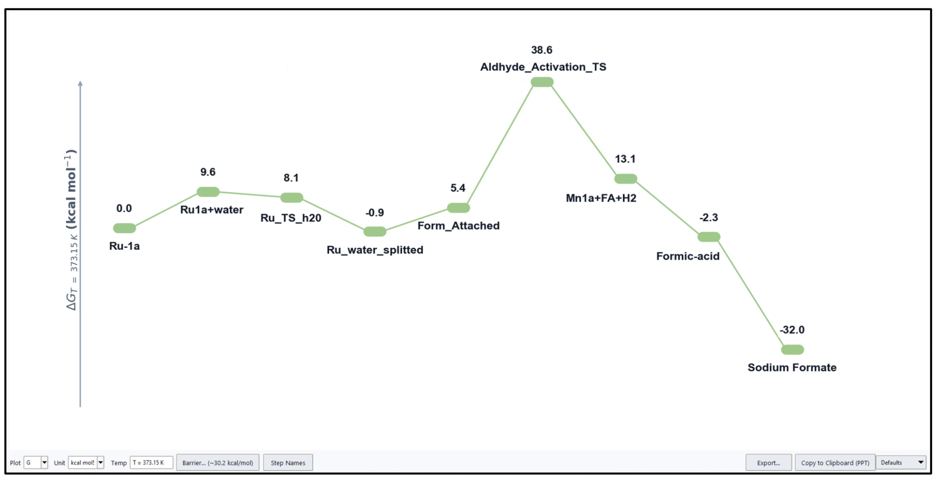
Publication-Ready Potential-Energy Plots

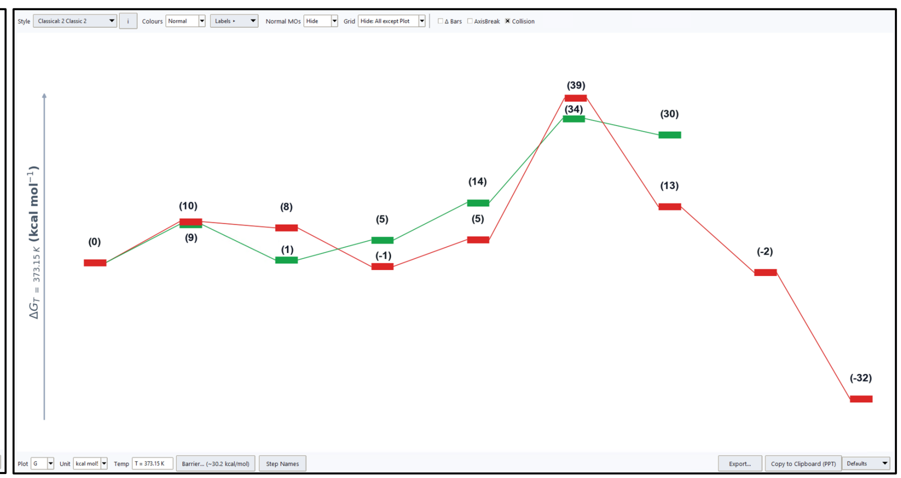
Generate a clean potential-energy profile from the selected pathway and refine the figure using built-in styles, labels, colors, grids, and axis controls.

Publication-quality plots for easy comparison of relative Gibbs free energies, enthalpies, and electronic energies of intermediates and transition states.









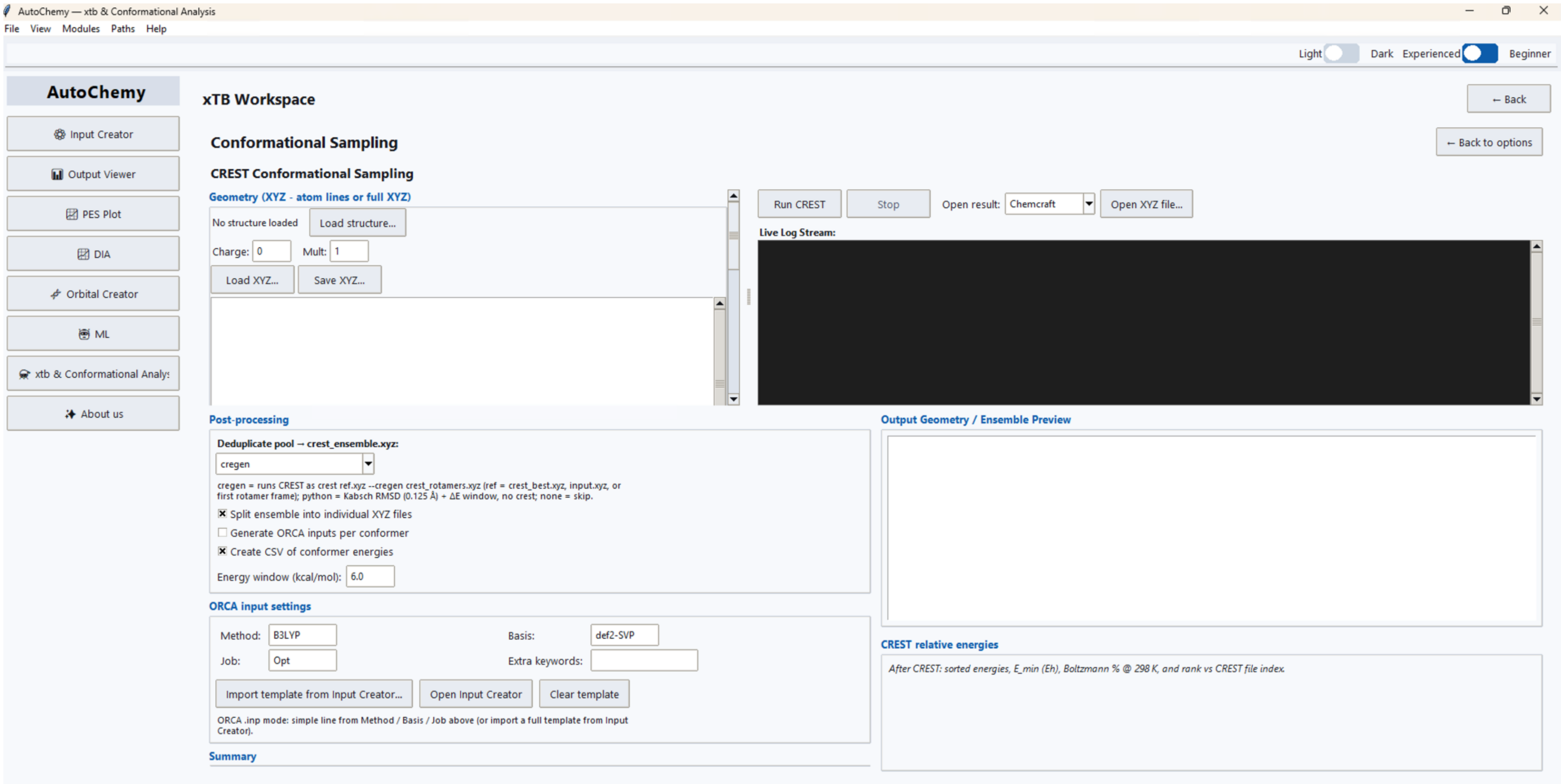
Scientific communication: Consistent energies and species labels turn calculation data into a figure that is ready for discussion or further polishing.

Rapid iteration: Style and visibility controls make it easy to compare figure treatments without rebuilding the plot manually.

FEATURE 09

CREST Conformational Sampling

Prepare and run CREST sampling, inspect logs and ensemble structures, deduplicate conformers, and create downstream ORCA inputs from selected geometries.



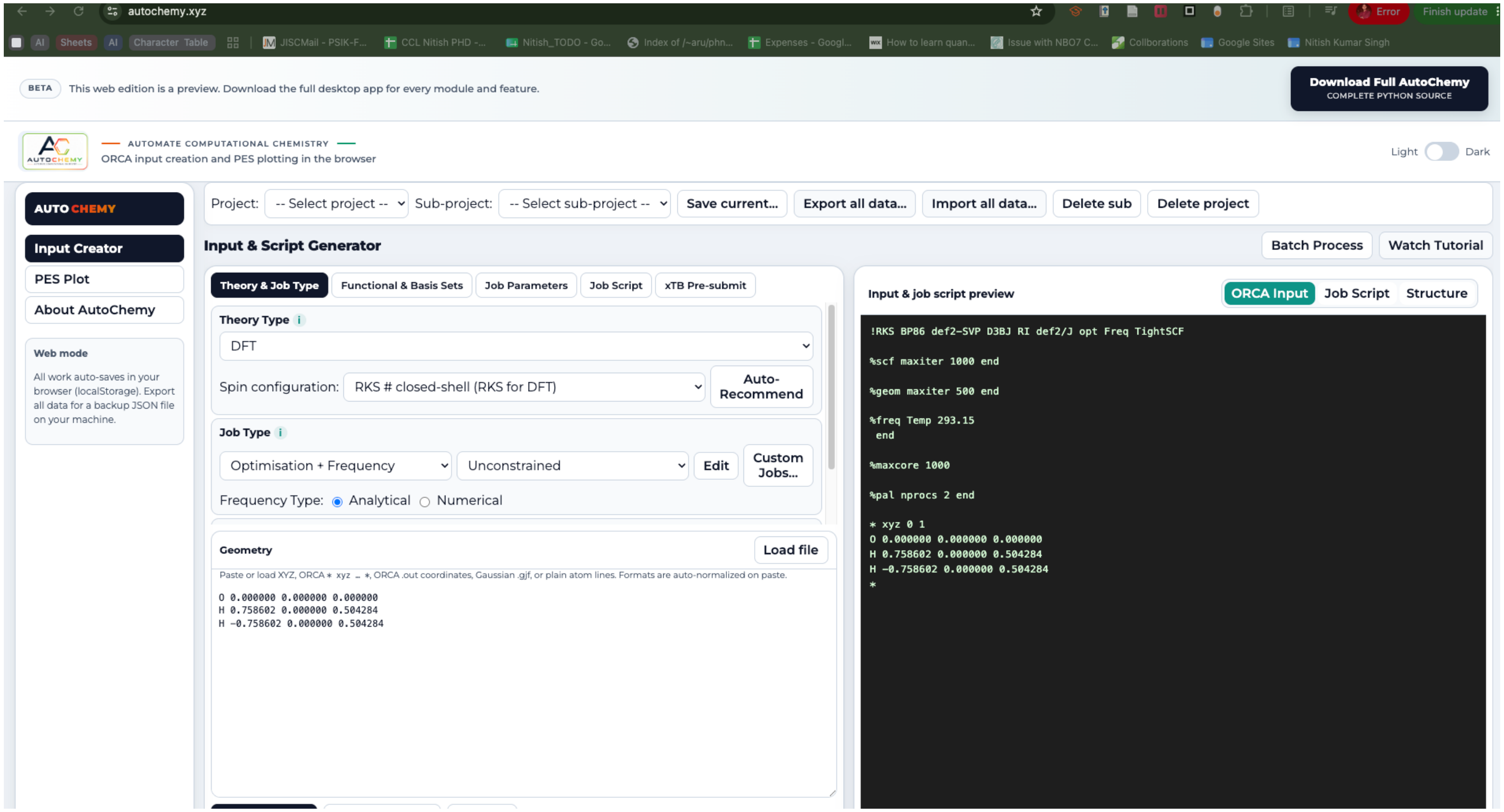
End-to-end sampling: Structure loading, charge and multiplicity, CREST execution, and post-processing are brought together.

Reusable outputs: Export individual XYZ files, conformer-energy summaries, and ORCA-ready inputs for follow-up calculations.

FEATURE 10

Live AutoChemy Web Beta

Open the AutoChemy beta directly at autochemy.xyz to create ORCA inputs and prepare PES workflows in a browser, with no desktop installation required for the preview modules.



Easy access: The beta is live in the browser, making it convenient to explore AutoChemy, prepare inputs, and demonstrate the interface from any suitable computer.

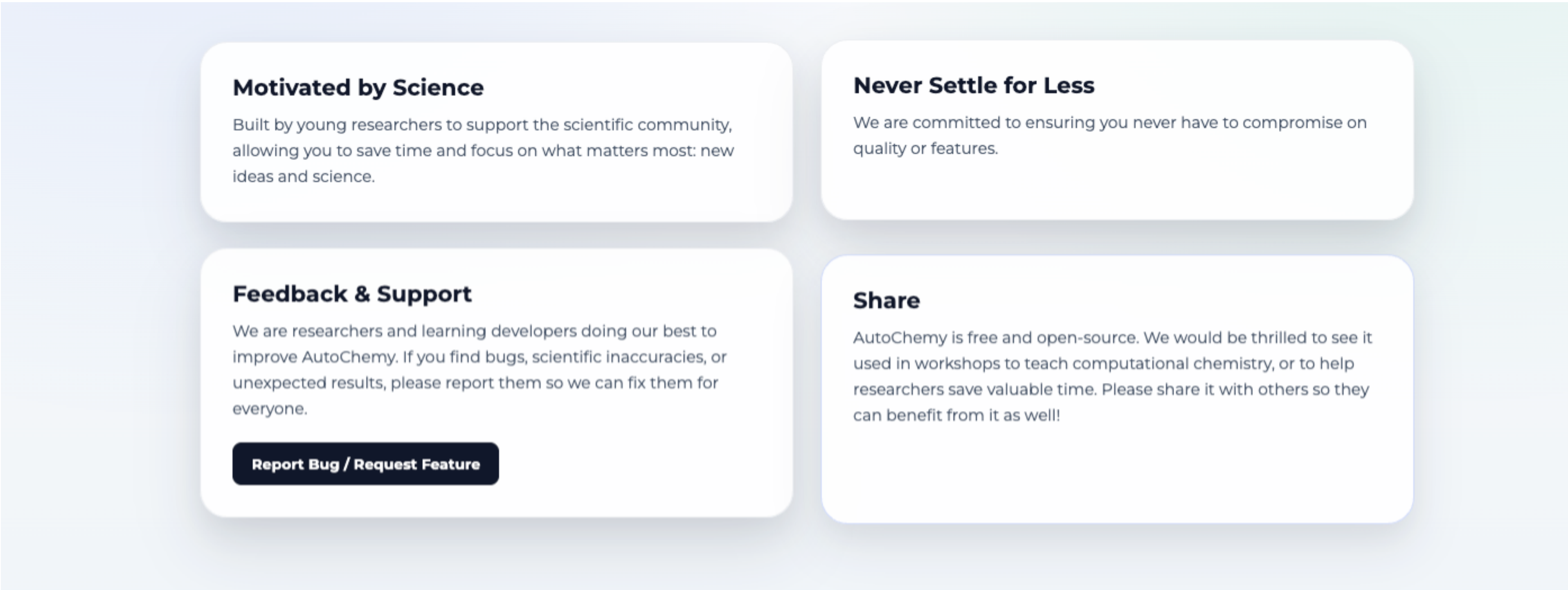
Browser-safe workflow: Projects auto-save locally in the browser and can be exported or imported as backup data; the complete desktop source remains available for all modules and features.

OPEN THE LIVE BETA autochemy.xyz

FEATURE 11

Open-Source and Researcher-Led

AutoChemy is developed by researchers to reduce repetitive quantum-chemistry work and help users spend more time on scientific interpretation.



Community use: The software is free to share for research, teaching, workshops, and computational-chemistry training.

Feedback loop: Users can report issues or request features, supporting continuous improvement of the scientific workflow.

EXPLORE AND DOWNLOAD autochemy.xyz | [Full source code on GitHub]()

FOR MORE DETAILS
CONTACT

NITISH KUMAR SINGH
Email: nitishdft@gmail.com